



SWE 101 : Cryptology
Computing Technologies Department

<Practical Report <1>>

**React Native Development Environment and Create a Basic
Multi-Screen Application**

<Submitted by>

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1. Aim

The goal of this practical is to set up the environment for React Native development and make a simple mobile app that has more than one screen and lets users move between them.

2. Objective

The goals of this exercise are:

- To set up and install the tools you need to make React Native apps
- To use Expo to make a new mobile app
- To learn how a React Native project is put together
- To make more than one screen in the app
- To make it possible to move between those screens

3. Learning Outcome

After completing this practical, the learner should be able to:

- Create and run a React Native app using Expo.
- Explain the purpose of core project files and folders.
- Build a simple application with at least two screens.
- Use basic navigation to move between screens.
- Write clean, modular, and readable React Native code.

4. Introduction

React Native is a framework that allows developers to build mobile applications using JavaScript. It is widely used because it allows the same code to run on both Android and iOS devices.

Expo is a tool that makes React Native easier to use, especially for beginners. It removes the need for complicated setup and allows the application to run directly on a mobile device using the Expo Go app.

In this practical, I created a simple application with two screens and implemented navigation between them.

5. Requirements

Software Requirements

- Node.js
- npm
- Expo Go app on a physical Android/iOS device, or an emulator if available
- Code editor such as Visual Studio Code

Hardware requirements

- A computer with internet access
- A smartphone for testing, if using Expo Go

6. Procedure

Step 1: Installing Node.js

First, I installed Node.js from the official website. After installation, I verified it by running:

node -v

npm -v

This confirmed that Node.js and npm were successfully installed.

Step 2: Creating the Expo Project

Next, I created a new React Native project using Expo by running:

npx create-expo-app Practical1App

Then I moved into the project folder:

cd Practical1App

After that, I started the project using:

npm start or npx expo start

This opened the Expo development server.

Step 3: Installing Navigation Packages

To enable navigation between screens, I installed the required packages:

@react-navigation/native

@react-navigation/native-stack

react-native-screens

react-native-safe-area-context

These packages are necessary for implementing stack navigation in the application.

Step 4: Creating Screens

I created two screens:

Home Screen

Second Screen

Each screen was placed in a separate file to keep the code organized and easy to manage.

Step 5: Setting Up Navigation

This app used stack navigation. I brought in the navigation parts I needed and made a stack navigator. I then put both screens inside the navigator so that the app knows how to switch between them.

Step 6: Adding Navigation Functionality

On the Home Screen, I added a button that takes the user to the Second Screen using:

```
navigation.navigate('Second')
```

On the Second Screen, I added a button to go back to the Home Screen using:

```
navigation.goBack()
```

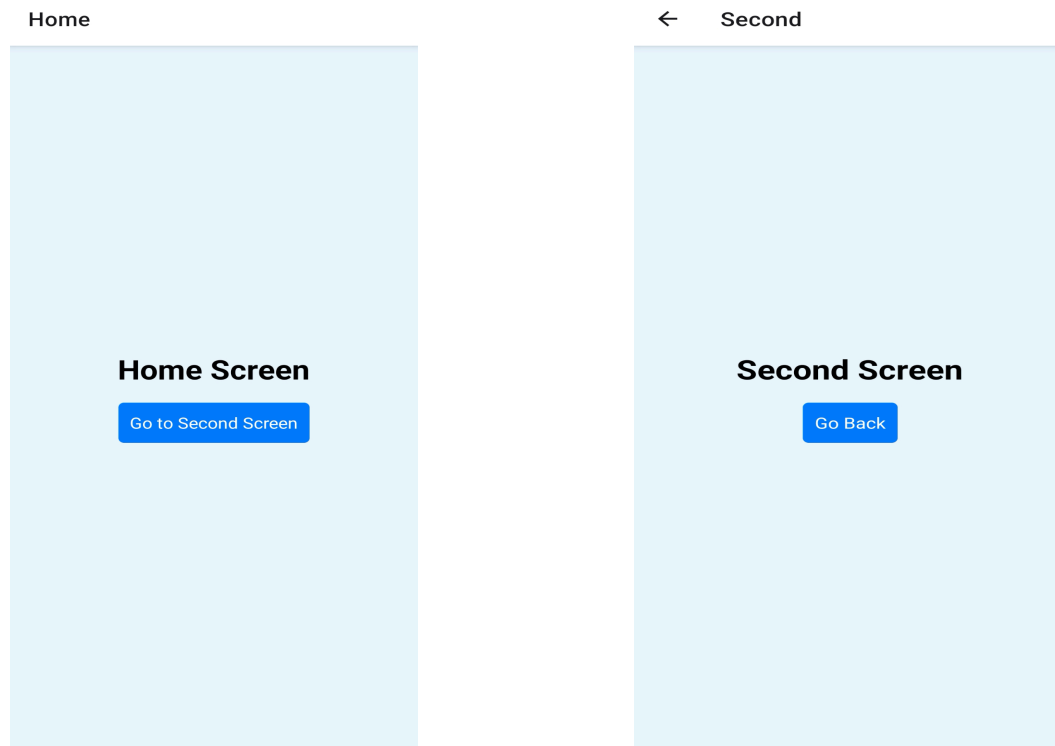
Step 7: Running and Testing the App

Finally, I tested the application using the Expo Go app on my mobile phone. I scanned the QR code and the app opened successfully. I checked whether navigation between screens was working properly.

7. Program / Code

Repository Link: https://github.com/Ldorji705/SWE_101_02240349.git

8. Output (In Image format).



9. Observation

During this practical, I observed that:

- Expo makes it very easy to run and test mobile applications
- Navigation works smoothly when all required packages are installed
- Even a small mistake in naming screens can cause errors
- Proper structure and organization of files makes the project easier to understand

10. Problem Encountered

I had a few small problems while doing the practical:

- At first, navigation didn't work because some dependencies were missing.
- I had to make sure that all the packages were set up right.
- I also had to be very careful when matching screen names in the navigation setup.

11. Conclusion

This practical was successfully completed. I was able to create a simple React Native application using Expo and implement navigation between two screens. This practical helped me understand the basics of mobile app development and improved my understanding of navigation in React Native.